15.2 LINE INTEGRALS

The first goal of this section is to define what it means to integrate a function along a curve. To motivate the definition we will consider the problem of finding the mass of a very thin wire whose linear density function (mass per unit length) is known. We assume that we can model the wire by a smooth curve C between two points P and Q in 3-space (Figure 15.2.1). Given any point (x, y, z) on C, we let $f(x, y, z)$ denote the corresponding value of the density function. To compute the mass of the wire, we proceed as follows:

- Divide C into n very small sections using a succession of distinct partition points

  $$P = P_0, P_1, P_2, \ldots, P_{n-1}, P_n = Q$$

  as illustrated on the left side of Figure 15.2.2. Let ΔM_k be the mass of the kth section, and let Δs_k be the length of the arc between P_{k-1} and P_k.

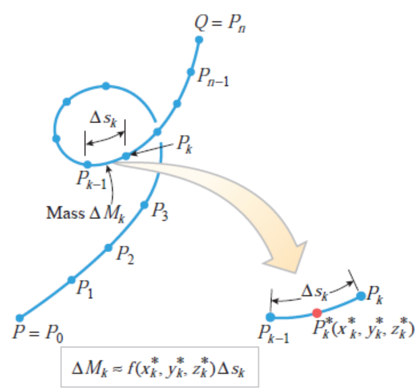


Figure 15.2.2

15.2.1 DEFINITION If C is a smooth curve in 2-space or 3-space, then the ***line integral of f with respect to s along C*** is

$$\int_C f(x, y)\, ds = \lim_{\max \Delta s_k \to 0} \sum_{k=1}^{n} f(x_k^*, y_k^*)\Delta s_k \qquad \text{2-space} \qquad (3)$$

or

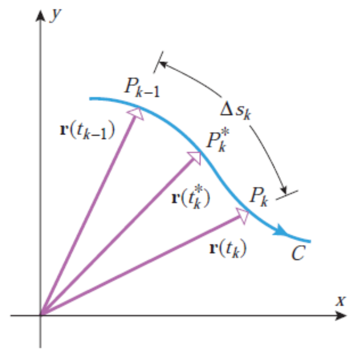
$$\int_C f(x, y, z)\, ds = \lim_{\max \Delta s_k \to 0} \sum_{k=1}^{n} f(x_k^*, y_k^*, z_k^*)\Delta s_k \qquad \text{3-space} \qquad (4)$$

provided this limit exists and does not depend on the choice of partition or on the choice of sample points.

EVALUATING LINE INTEGRALS

Except in simple cases, it will not be feasible to evaluate a line integral directly from (3) or (4). However, we will now show that it is possible to express a line integral as an ordinary definite integral, so that no special methods of evaluation are required. For example, suppose that C is a curve in the xy-plane that is smoothly parametrized by

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} \qquad (a \le t \le b)$$



▲ **Figure 15.2.5**

Therefore, if C is smoothly parametrized by

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} \qquad (a \le t \le b)$$

then

$$\int_C f(x, y)\, ds = \int_a^b f(x(t), y(t))\|\mathbf{r}'(t)\|\, dt \tag{9}$$

Similarly, if C is a curve in 3-space that is smoothly parametrized by

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k} \qquad (a \le t \le b)$$

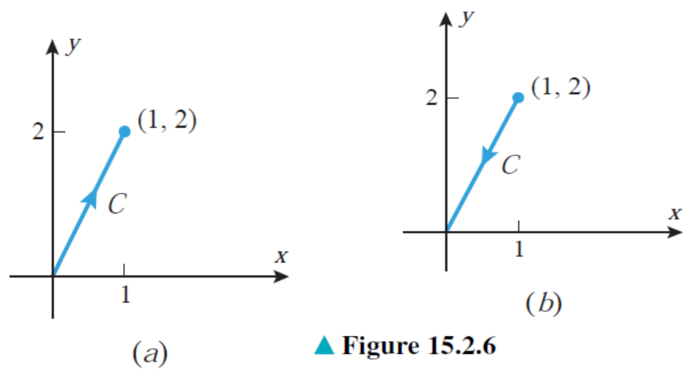
then

$$\int_C f(x, y, z)\, ds = \int_a^b f(x(t), y(t), z(t))\|\mathbf{r}'(t)\|\, dt \tag{10}$$

▶ **Example 1** Using the given parametrization, evaluate the line integral $\int_C (1 + xy^2)\, ds$.

(a) $C : \mathbf{r}(t) = t\mathbf{i} + 2t\mathbf{j} \quad (0 \le t \le 1)$ (see Figure 15.2.6*a*)

(b) $C : \mathbf{r}(t) = (1 - t)\mathbf{i} + (2 - 2t)\mathbf{j} \quad (0 \le t \le 1)$ (see Figure 15.2.6*b*)



▲ **Figure 15.2.6**

Solution (a). Since $\mathbf{r}'(t) = \mathbf{i} + 2\mathbf{j}$, we have $\|\mathbf{r}'(t)\| = \sqrt{5}$ and it follows from Formula (9) that

$$\begin{aligned}\int_C (1 + xy^2)\,ds &= \int_0^1 [1 + t(2t)^2]\sqrt{5}\,dt \\ &= \int_0^1 (1 + 4t^3)\sqrt{5}\,dt \\ &= \sqrt{5}\left[t + t^4\right]_0^1 = 2\sqrt{5}\end{aligned}$$

Solution (b). Since $\mathbf{r}'(t) = -\mathbf{i} - 2\mathbf{j}$, we have $\|\mathbf{r}'(t)\| = \sqrt{5}$ and it follows from Formula (9) that

$$\begin{aligned}\int_C (1 + xy^2)\,ds &= \int_0^1 [1 + (1-t)(2-2t)^2]\sqrt{5}\,dt \\ &= \int_0^1 [1 + 4(1-t)^3]\sqrt{5}\,dt \\ &= \sqrt{5}\left[t - (1-t)^4\right]_0^1 = 2\sqrt{5}\end{aligned}$$ ◀

Formula (9) has an alternative expression for a curve C in the xy-plane that is given by parametric equations

$$x = x(t), \quad y = y(t) \qquad (a \le t \le b)$$

In this case, we write (9) in the expanded form

$$\int_C f(x, y)\, ds = \int_a^b f(x(t), y(t))\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \tag{11}$$

Similarly, if C is a curve in 3-space that is parametrized by

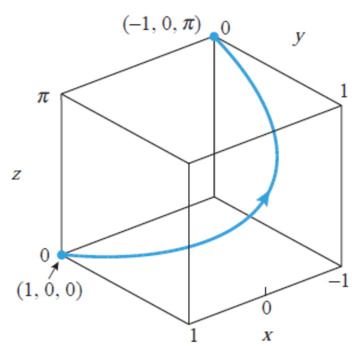
$$x = x(t), \quad y = y(t), \quad z = z(t) \qquad (a \le t \le b)$$

then we write (10) in the form

$$\int_C f(x, y, z)\, ds = \int_a^b f(x(t), y(t), z(t))\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt \tag{12}$$

▶ **Example 2** Evaluate the line integral $\int_C (xy + z^3)\, ds$ from $(1, 0, 0)$ to $(-1, 0, \pi)$ along the helix C that is represented by the parametric equations

$$x = \cos t, \quad y = \sin t, \quad z = t \qquad (0 \le t \le \pi)$$



▲ **Figure 15.2.7**

Solution. From (12)

$$\begin{aligned}
\int_C (xy + z^3)\, ds &= \int_0^\pi (\cos t \sin t + t^3)\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt \\
&= \int_0^\pi (\cos t \sin t + t^3)\sqrt{(-\sin t)^2 + (\cos t)^2 + 1}\, dt \\
&= \sqrt{2}\int_0^\pi (\cos t \sin t + t^3)\, dt \\
&= \sqrt{2}\left[\frac{\sin^2 t}{2} + \frac{t^4}{4}\right]_0^\pi = \frac{\sqrt{2}\pi^4}{4}
\end{aligned}$$ ◀

- If C is a curve in 3-space that models a thin wire, and if $f(x, y, z)$ is the linear density function of the wire, then it follows from (2) and Definition 15.2.1 that the mass M of the wire is given by

$$M = \int_C f(x, y, z)\, ds \tag{5}$$

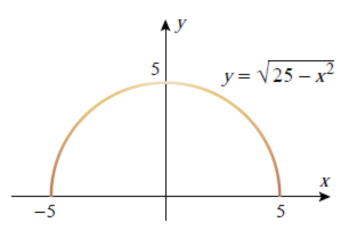
  That is, to obtain the mass of a thin wire, we integrate the linear density function over the smooth curve that models the wire.

It is then plausible that

$$\text{Area,} \qquad A = \lim_{\max \Delta s_k \to 0} \sum_{k=1}^{n} f(x_k^*, y_k^*)\Delta s_k = \int_C f(x, y)\, ds \tag{7}$$

If $\delta(x, y)$ is the linear density function of a wire that is modeled by a smooth curve C in the xy-plane, then an argument similar to the derivation of Formula (5) shows that the mass of the wire is given by $\int_C \delta(x, y)\, ds$.

▶ **Example 3** Suppose that a semicircular wire has the equation $y = \sqrt{25 - x^2}$ and that its mass density is $\delta(x, y) = 15 - y$ (Figure 15.2.8). Physically, this means the wire has a maximum density of 15 units at the base ($y = 0$) and that the density of the wire decreases linearly with respect to y to a value of 10 units at the top ($y = 5$). Find the mass of the wire.



▲ **Figure 15.2.8**

Solution. The mass M of the wire can be expressed as the line integral

$$M = \int_C \delta(x, y)\, ds = \int_C (15 - y)\, ds$$

along the semicircle C. To evaluate this integral we will express C parametrically as

$$x = 5\cos t, \quad y = 5\sin t \qquad (0 \le t \le \pi)$$

Thus, it follows from (11) that

$$M = \int_C (15 - y)\,ds = \int_0^{\pi} (15 - 5\sin t)\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$$
$$= \int_0^{\pi} (15 - 5\sin t)\sqrt{(-5\sin t)^2 + (5\cos t)^2}\,dt$$
$$= 5\int_0^{\pi} (15 - 5\sin t)\,dt$$
$$= 5\left[15t + 5\cos t\right]_0^{\pi}$$
$$= 75\pi - 50 \approx 185.6 \text{ units of mass} \blacktriangleleft$$

Note: Radicals r=||r(t)||=1 in special case.

In the special case where t is an arc length parameter, say $t = s$, it follows that the radicals in (11) and (12) reduce to 1 and the equations simplify to

$$\int_C f(x, y)\,ds = \int_a^b f(x(s), y(s))\,ds \tag{13}$$

and

$$\int_C f(x, y, z)\,ds = \int_a^b f(x(s), y(s), z(s))\,ds \tag{14}$$

respectively.

Example 4 Find the area of the surface extending upward from the circle $x^2 + y^2 = 1$ in the xy-plane to the parabolic cylinder $z = 1 - x^2$ (Figure 15.2.9).

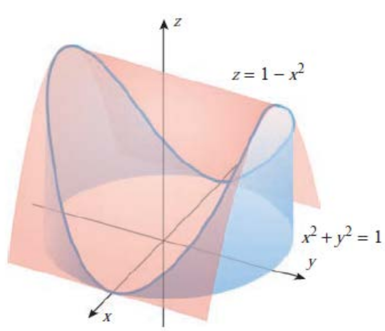


Figure 15.2.9

Solution. It follows from (7) that the area A of the surface can be expressed as the line integral

$$A = \int_C (1 - x^2)\, ds \tag{15}$$

where C is the circle $x^2 + y^2 = 1$. This circle can be parametrized in terms of arc length as

$$x = \cos s, \quad y = \sin s \qquad (0 \leq s \leq 2\pi)$$

Thus, it follows from (13) and (15) that

$$\begin{aligned} A &= \int_C (1 - x^2)\, ds = \int_0^{2\pi} (1 - \cos^2 s)\, ds \\ &= \int_0^{2\pi} \sin^2 s\, ds = \frac{1}{2}\int_0^{2\pi} (1 - \cos 2s)\, ds = \pi \end{aligned}$$

◀

LINE INTEGRALS WITH RESPECT TO *x*, *y*, AND *z*

The basic procedure for evaluating these line integrals is to find parametric equations for C, say

$$x = x(t), \quad y = y(t), \quad z = z(t) \qquad (a \leq t \leq b)$$

in which the orientation of C is in the direction of increasing t, and then express the integrand in terms of t. For example,

$$\boxed{\int_C f(x, y, z)\, dz = \int_a^b f(x(t), y(t), z(t))z'(t)\, dt}$$

[Such a formula is easy to remember—just substitute for x, y, and z using the parametric equations and recall that $dz = z'(t)\, dt$.]

Example 5 Evaluate $\int_C 3xy\,dy$, where C is the line segment joining $(0, 0)$ and $(1, 2)$ with the given orientation.

(a) Oriented from $(0, 0)$ to $(1, 2)$ as in Figure 15.2.6a.

(b) Oriented from $(1, 2)$ to $(0, 0)$ as in Figure 15.2.6b.

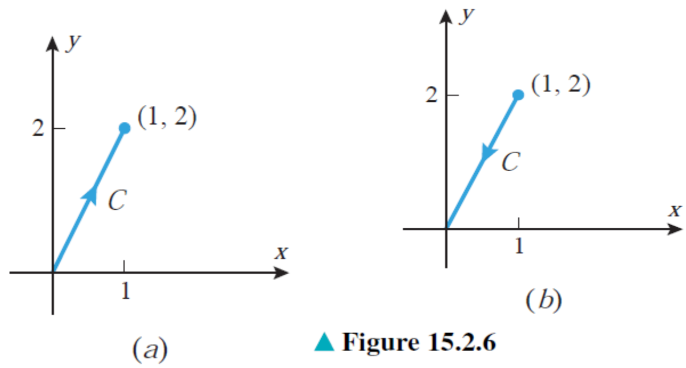


Figure 15.2.6

Solution (a). Using the parametrization

$$x = t, \quad y = 2t \qquad (0 \le t \le 1)$$

we have

$$\int_C 3xy\,dy = \int_0^1 3(t)(2t)(2)\,dt = \int_0^1 12t^2\,dt = 4t^3\Big]_0^1 = 4$$

Solution (b). Using the parametrization

$$x = 1 - t, \quad y = 2 - 2t \qquad (0 \le t \le 1)$$

we have

$$\int_C 3xy\,dy = \int_0^1 3(1-t)(2-2t)(-2)\,dt = \int_0^1 -12(1-t)^2\,dt = 4(1-t)^3\Big]_0^1 = -4$$

Frequently, the line integrals with respect to x and y occur in combination, in which case we will dispense with one of the integral signs and write

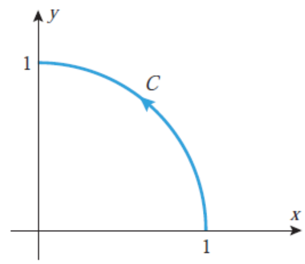
$$\int_C f(x, y)\,dx + g(x, y)\,dy = \int_C f(x, y)\,dx + \int_C g(x, y)\,dy \tag{21}$$

We will use a similar convention for combinations of line integrals with respect to x, y, and z along curves in 3-space.

▶ **Example 6** Evaluate

$$\int_C 2xy\,dx + (x^2 + y^2)\,dy$$

along the circular arc C given by $x = \cos t$, $y = \sin t$ $(0 \le t \le \pi/2)$ (Figure 15.2.11).



$x = \cos t,\ y = \sin t\ (0 \le t \le \pi/2)$

▲ **Figure 15.2.11**

Solution. We have

$$\begin{aligned}
\int_C 2xy\,dx &= \int_0^{\pi/2} (2\cos t \sin t)\left[\frac{d}{dt}(\cos t)\right] dt \\
&= -2\int_0^{\pi/2} \sin^2 t \cos t\,dt = -\frac{2}{3}\sin^3 t\Big]_0^{\pi/2} = -\frac{2}{3} \\
\int_C (x^2 + y^2)\,dy &= \int_0^{\pi/2} (\cos^2 t + \sin^2 t)\left[\frac{d}{dt}(\sin t)\right] dt \\
&= \int_0^{\pi/2} \cos t\,dt = \sin t\Big]_0^{\pi/2} = 1
\end{aligned}$$

Thus, from (21)

$$\begin{aligned}
\int_C 2xy\,dx + (x^2 + y^2)\,dy &= \int_C 2xy\,dx + \int_C (x^2 + y^2)\,dy \\
&= -\frac{2}{3} + 1 = \frac{1}{3} \quad ◀
\end{aligned}$$

INTEGRATING A VECTOR FIELD ALONG A CURVE

There is an alternative notation for line integrals with respect to x, y, and z that is particularly appropriate for dealing with problems involving vector fields. We will interpret $d\mathbf{r}$ as

$$d\mathbf{r} = dx\,\mathbf{i} + dy\,\mathbf{j} \quad \text{or} \quad d\mathbf{r} = dx\,\mathbf{i} + dy\,\mathbf{j} + dz\,\mathbf{k}$$

depending on whether C is in 2-space or 3-space. For an oriented curve C in 2-space and a vector field

$$\mathbf{F}(x, y) = f(x, y)\mathbf{i} + g(x, y)\mathbf{j}$$

we will write

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (f(x, y)\mathbf{i} + g(x, y)\mathbf{j}) \cdot (dx\,\mathbf{i} + dy\,\mathbf{j}) = \int_C f(x, y)\,dx + g(x, y)\,dy \tag{26}$$

Similarly, for a curve C in 3-space and vector field

$$\mathbf{F}(x, y, z) = f(x, y, z)\mathbf{i} + g(x, y, z)\mathbf{j} + h(x, y, z)\mathbf{k}$$

we will write

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C (f(x, y, z)\mathbf{i} + g(x, y, z)\mathbf{j} + h(x, y, z)\mathbf{k}) \cdot (dx\,\mathbf{i} + dy\,\mathbf{j} + dz\,\mathbf{k}) \\ &= \int_C f(x, y, z)\,dx + g(x, y, z)\,dy + h(x, y, z)\,dz\end{aligned} \tag{27}$$

15.2.2 DEFINITION If $\mathbf{F}$ is a continuous vector field and C is a smooth oriented curve, then the ***line integral of* F *along C*** is

$$\int_C \mathbf{F} \cdot d\mathbf{r} \tag{28}$$

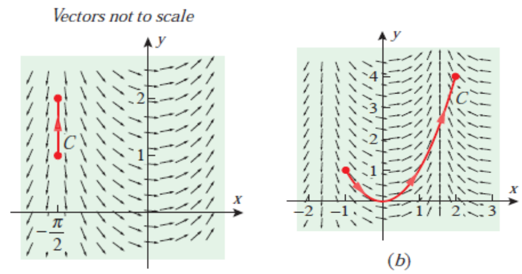
Finally after simplification:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\,dt \tag{29}$$

▶ Example 8 Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y) = \cos x\,\mathbf{i} + \sin x\,\mathbf{j}$ and where C is the given oriented curve.

(a) $C : \mathbf{r}(t) = -\dfrac{\pi}{2}\mathbf{i} + t\mathbf{j} \quad (1 \le t \le 2)$ (see Figure 15.2.12a)

(b) $C : \mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} \quad (-1 \le t \le 2)$ (see Figure 15.2.12b)



(a) (b)

▲ Figure 15.2.12

Solution (a). Using (29) we have

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_1^2 \mathbf{F}(\mathbf{r}(t))\cdot \mathbf{r}'(t)\,dt = \int_1^2 (-\mathbf{j})\cdot \mathbf{j}\,dt = \int_1^2 (-1)\,dt = -1$$

Solution (b). Using (29) we have

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_{-1}^2 \mathbf{F}(\mathbf{r}(t))\cdot \mathbf{r}'(t)\,dt = \int_{-1}^2 (\cos t\,\mathbf{i} + \sin t\,\mathbf{j})\cdot(\mathbf{i} + 2t\mathbf{j})\,dt$$

$$= \int_{-1}^2 (\cos t + 2t\sin t)\,dt = (-2t\cos t + 3\sin t)\Big]_{-1}^2$$

$$= -2\cos 1 - 4\cos 2 + 3(\sin 1 + \sin 2) \approx 5.83629 \blacktriangleleft$$

Practice Problem:

Chapter-15.2:

Problems: 11, 13, 14, 21-27, 45-48

Thank you for your attention!